

COMPTIA N+ ASSESSMENT

Network Security, Maintenance & Troubleshooting Procedures



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SCENARIO – 1

Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

Q - What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

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1. CPU (Central Processing Unit)

- **Speed: Choose a fast CPU so that programs run smoothly.**
- **Cores: A CPU with multiple cores helps run different applications at the same time.**
- **Compatibility: The CPU must be compatible with the motherboard.**

2. RAM (Random Access Memory)

- **Capacity: Choose at least 16 GB RAM to run programming software, web browsers, and video calls together.**
- **Type: Check whether the motherboard supports DDR4 or DDR5 RAM.**
- **Upgrade: Choose a computer that allows RAM to be upgraded in the future.**

3. Storage

- **Type: Choose an SSD because it loads files and applications faster than an HDD.**
- **Capacity: At least 512 GB storage is recommended for software and files.**
- **Compatibility: Check whether the motherboard supports the selected storage device.**

SCENARIO – 2

After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

Q - Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

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1. CPU (Central Processing Unit)

- **Antivirus software uses CPU power to scan files and check for viruses.**
- **When scanning runs in the background, other programs may run slowly.**

2. RAM (Random Access Memory)

- **Antivirus software uses RAM to run its processes.**
- **If RAM is limited, less memory is available for other applications, making the system slower.**

3. Storage

- **Antivirus software uses storage space to install the program and save updates and logs.**
- **During scanning, it reads many files from the storage, which can slow down other tasks, especially on an HDD.**

SCENARIO – 3

A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Q - Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

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Steps to Troubleshoot the Problem

1. Check USB Connections

- **Make sure the keyboard and printer are connected properly.**
- **Try connecting them to a different USB port.**

2. Check Device Manager

- **Open Device Manager on the computer.**
- **Check whether the keyboard or printer shows a yellow warning sign.**
- **If there is a warning, the driver may have a problem.**

3. Update Drivers

- **Open Device Manager and select the faulty device.**
- **Right-click the device and select Update driver.**
- **Install the correct driver for the device.**

4. Reinstall Drivers

- **If updating does not work, uninstall the faulty driver.**
- **Restart the computer and reinstall the correct driver.**

5. Check the Operating System

- **Make sure the operating system supports the keyboard and printer.**
- **Install the latest Windows updates if required.**

6. Restart and Test

- **Restart the computer.**
- **Reconnect the devices and check whether they work correctly.**

SCENARIO – 4

Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Q - Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

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1. UPS (Uninterruptible Power Supply)

- **Backup Power: A UPS provides temporary power during a power cut.**
- **Prevent Data Loss: It gives employees time to save their work and shut down computers safely.**
- **Protect Hardware: It helps protect computers from sudden power interruptions.**

2. Surge Protector

- **Voltage Protection: It protects computer parts from sudden increases in voltage.**
- **Prevent Damage: It helps prevent damage to the motherboard, CPU, and other components.**
- **Safe Power Supply: It reduces the risk of damage caused by voltage spikes.**

SCENARIO – 5

New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.

Q - Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described

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1. DNS (Domain Name System)

- **DNS converts a domain name into an IP address.**
- **For example, it converts server.company.com into the server's IP address.**
- **If DNS is not working properly, employees can access the server by IP address but not by name.**

2. DHCP (Dynamic Host Configuration Protocol)

- **DHCP automatically assigns IP addresses to computers.**
- **It also provides the default gateway, subnet mask, and DNS server address.**
- **If DHCP is not working properly, computers may fail to obtain a valid IP address.**

3. How DNS and DHCP Work Together

- **DHCP provides the computer with an IP address and DNS server details.**
- **DNS uses name-to-IP address conversion to help the computer find internal systems.**
- **Both services help employees access network resources easily.**

4. Which Service Is Misconfigured?

- **DNS issue: Employees can access the server by IP address but not by name. This suggests a DNS configuration problem.**
- **DHCP issue: Devices sometimes fail to obtain a valid IP address. This suggests a DHCP configuration problem.**

SCENARIO – 6

After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

Q - Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

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Steps to Troubleshoot the Problem

1. Check Network Connectivity

- **Check whether employees can connect to the network.**
- **Try pinging the application server using its IP address.**

2. Check Firewall Rules

- **Open the firewall settings.**
- **Check whether any rule is blocking traffic to the application server.**
- **Verify the server's IP address and the required ports.**

3. Check Firewall Logs

- **Review firewall logs to identify blocked connections.**
- **Check which rule is blocking employees from accessing the server.**

4. Modify Firewall Rules

- **Create or update a rule to allow legitimate traffic to the application server.**
- **Allow only the required IP addresses, ports, and protocols.**
- **Avoid disabling the firewall completely.**

5. Test the Connection

- **Ask employees to access the application server again.**
- **Confirm that the server works properly and that other firewall protections remain active.**

Section B — Practical Application

Task 1: Workstation Assembly

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1. Install Components: Install the CPU, CPU cooler, RAM, and SSD on the motherboard in the correct way.

2. Assemble the Chassis: Place the motherboard inside the cabinet and secure it with screws. Install the power supply (PSU).

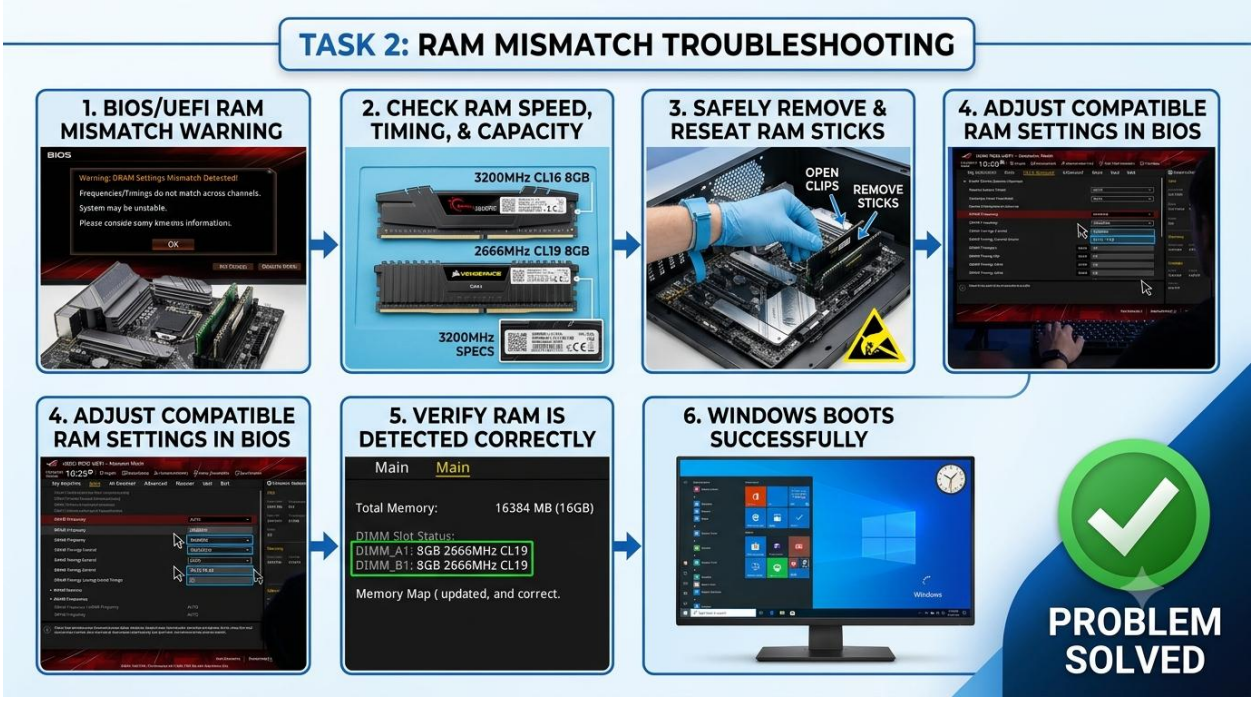
3. Connect Cables: Connect the motherboard power cable, CPU power cable, storage data cable, and front-panel cables.

4. Cable Management: Arrange the cables neatly and check that all connections are secure. Close the cabinet.

5. Power On and Test: Connect the monitor, keyboard, and mouse. Turn on the computer and check whether it passes POST. Verify that the CPU, RAM, and storage are detected in BIOS/UEFI.

Task 2: RAM Mismatch Troubleshooting

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1. Identify the Error: Start the computer and check the BIOS/UEFI POST message for a RAM mismatch warning.

2. Check the Cause: Check whether the RAM modules have different speeds, timings, or capacities.

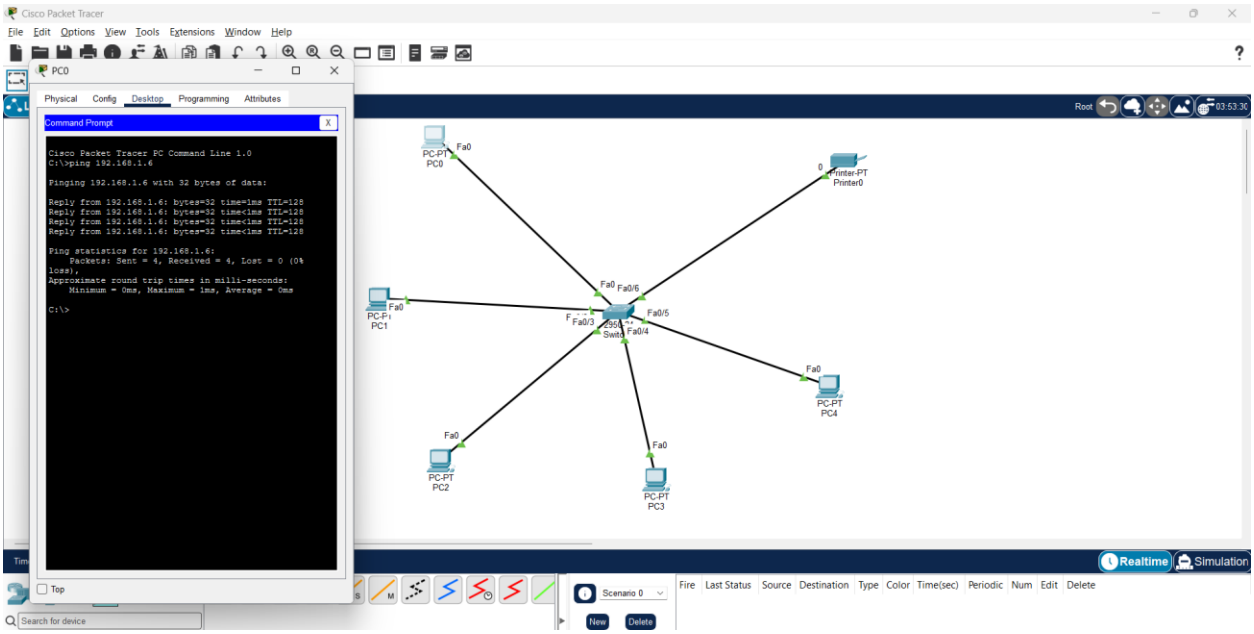
3. Reseat the RAM: Turn off the computer and disconnect the power. Remove the RAM sticks and reinstall them properly in the correct slots.

4. Adjust BIOS Settings: Open BIOS/UEFI and set the memory speed and timings to compatible settings, such as Auto. Save the changes and restart.

5. Test the System: Check whether the BIOS detects the RAM correctly and whether the computer boots normally into Windows.

Task 3: LAN Topology Build

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Devices Required: 1 switch, 5 PCs, 1 printer, and Ethernet cables.

IP Address Configuration:

Device	IP Address
PC0	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4
PC4	192.168.1.5
Printer	192.168.1.6

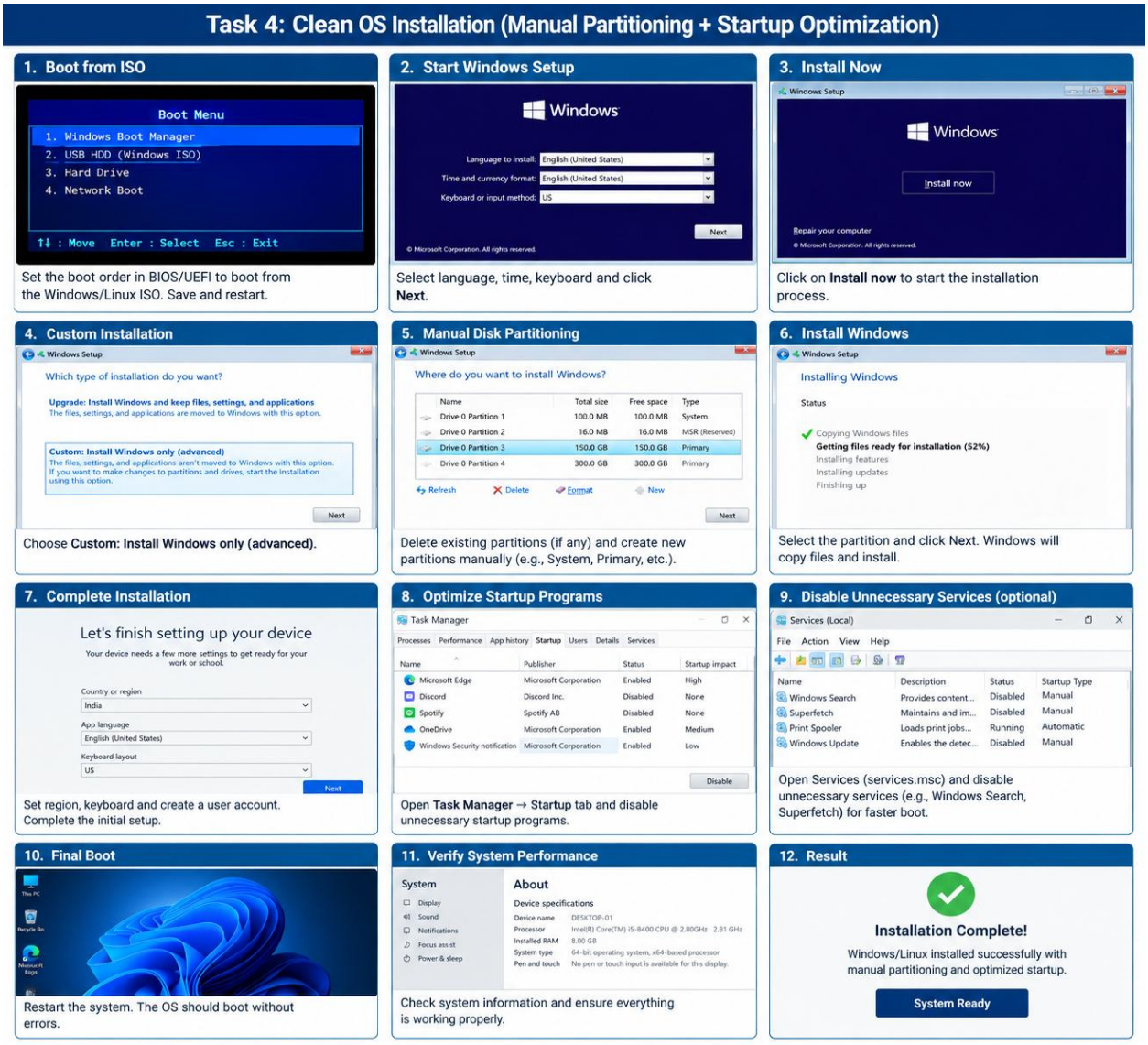
Subnet Mask: 255.255.255.0

Procedure:

1. **Connect all PCs and the printer to the switch.**
2. **Assign the IP addresses and subnet mask to each device.**
3. **Use the ping command to test connectivity between devices.**

Task 4: Clean OS Installation

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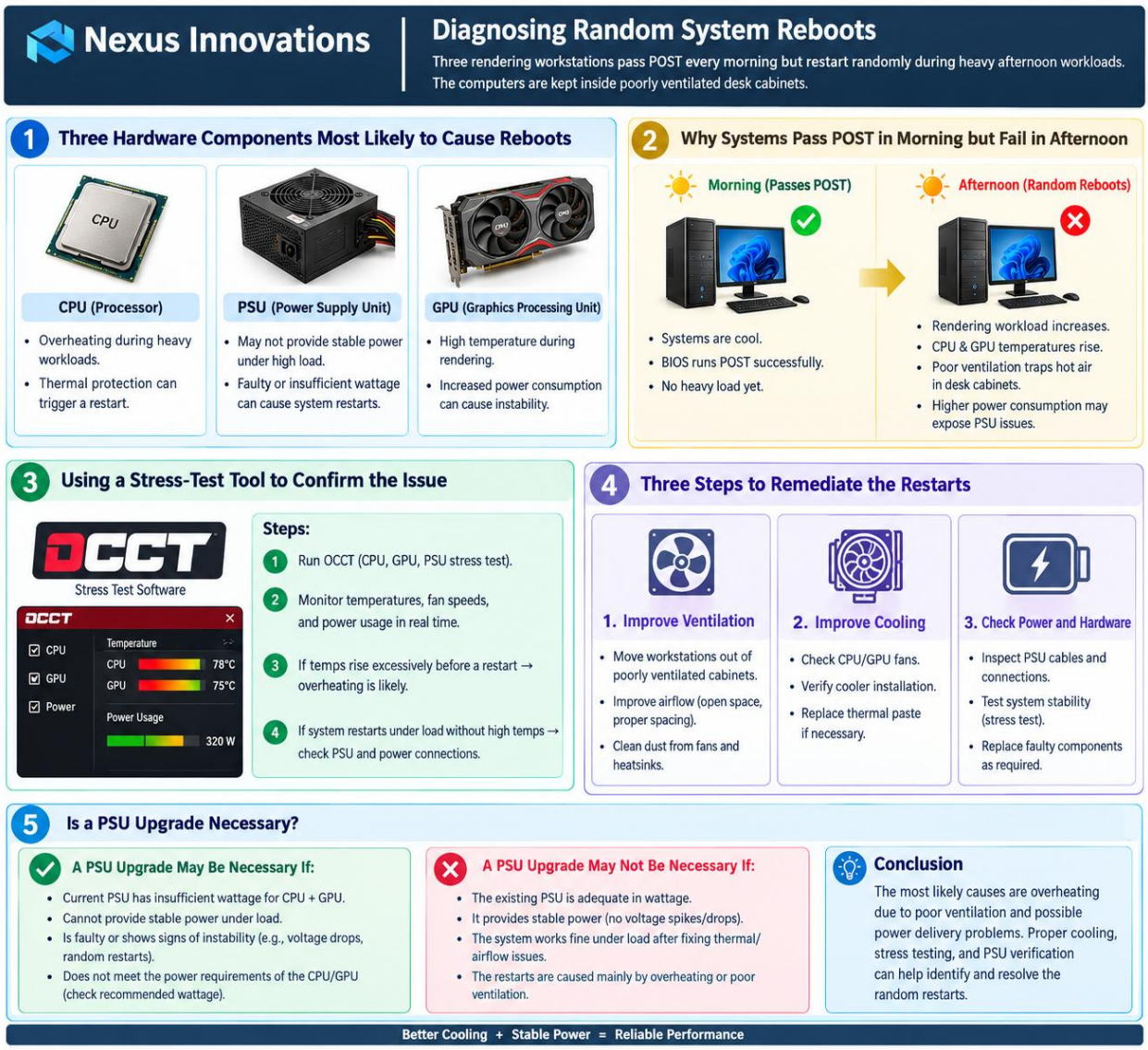


1. **Create a bootable USB using the Windows/Linux ISO.**
2. **Boot the workstation from the USB and start the OS installation.**
3. **Manually create disk partitions, such as system, OS, and data partitions. Back up important data before deleting or formatting partitions.**
4. **Complete the installation and install the required drivers.**
5. **Disable unnecessary startup applications and services.**
6. **Restart the workstation and check that the OS boots without errors.4**

Section C — Mini Capstone Project

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

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1. Three Hardware Components Most Likely to Cause Random Reboots

1. **CPU (Processor): High temperatures during heavy workloads can cause overheating and automatic system shutdowns or restarts.**
2. **PSU (Power Supply Unit): An insufficient or faulty PSU may fail to provide stable power when the CPU and GPU consume more electricity.**
3. **GPU (Graphics Processing Unit): Heavy rendering increases GPU temperature and power consumption, which can cause instability or system restarts.**

2. Why Systems Pass POST in the Morning but Fail in the Afternoon

In the morning, the systems are cool and pass POST (Power-On Self-Test) successfully. During afternoon rendering workloads, CPU and GPU temperatures rise. Poor ventilation inside desk cabinets traps hot air, causing overheating. Increased power consumption under load may also expose PSU problems, resulting in random restarts.

3. Using a Stress-Test Tool

- **Use a tool such as OCCT to test CPU, GPU, and power stability.**
- **Monitor CPU/GPU temperatures, fan speeds, and power usage during the test.**
- **If temperatures rise excessively before a restart, overheating is a likely cause.**
- **If the system restarts under combined CPU/GPU load without excessive temperatures, investigate the PSU and power connections.**

4. Three Steps to Remediate the Restarts

1. **Improve ventilation: Move workstations out of poorly ventilated cabinets, improve airflow, and clean dust from fans and heatsinks.**
2. **Improve cooling: Check CPU/GPU fans, verify cooler installation, and replace thermal paste if necessary.**
3. **Check power and hardware: Inspect PSU cables and connections, test system stability, and replace faulty components as required.**

5. Is a PSU Upgrade Necessary?

A PSU upgrade is necessary only if the existing PSU has insufficient wattage, cannot provide stable power, is faulty, or does not meet the CPU/GPU power requirements. If the PSU is adequate and stable, upgrading it may not solve the problem. Thermal and airflow issues should be addressed first based on test results.

Section D — AI-Augmented Learning

Step 1: Build with AI

1. Exact Prompt Used

Create an IP addressing and subnetting plan for Nexus Innovations' startup office network with 5 computers (PC1, PC2, PC3, PC4, PC5) and 1 network printer. Specify the network address, subnet mask, usable host range, broadcast address, and IP address for each device. Reserve some IP addresses for future expansion. Use a private IPv4 address range and explain the plan simply.

2. AI's Original Subnetting Plan

Network information

Parameter	Value
Network address	192.168.1.0/28
Subnet mask	255.255.255.240
Usable host range	192.168.1.1 – 192.168.1.14
Broadcast address	192.168.1.15
Total usable IP addresses	14

IP address allocation

Device	IP address
Router (Default Gateway)	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4

PC4	**192.168.1.5**
PC5	**192.168.1.6**
Printer	**192.168.1.7**
Reserved for future expansion	**192.168.1.8 – 192.168.1.14**

Step 2: Test and Debug (Without AI)

3. Verify and Correct the Plan

Step 1: Check the subnet mask

The /28 prefix corresponds to 255.255.255.240.

Step 2: Calculate the block size

Block size = 256 − 240 = 16.

Step 3: Verify the network and broadcast addresses

- **Network address: 192.168.1.0**
- **Broadcast address: 192.168.1.15**

Step 4: Verify the usable host range

The usable range is 192.168.1.1 to 192.168.1.14.

Step 5: Check capacity and future expansion

- **Total usable addresses: 14**
- **Addresses required for the router, five computers, and printer: 7**
- **Addresses reserved for future expansion: 7**
- **Overlapping IP addresses: None**

Corrected plan: The original AI-generated plan is correct. No changes are required because the subnet has sufficient capacity and all addresses are valid.

4. Corrected IP Addressing Plan

Device	Corrected IP address
Router / Default Gateway	192.168.1.1
PC1	192.168.1.2
PC2	192.168.1.3
PC3	192.168.1.4
PC4	192.168.1.5
PC5	192.168.1.6
Network Printer	192.168.1.7
Future expansion	192.168.1.8 – 192.168.1.14

Step 3: Note Explaining Changes (3–4 Lines)

I checked the AI-generated plan for overlapping IP addresses, incorrect network and broadcast addresses, and insufficient host capacity. The original plan was correct, so no changes were necessary. The /28 subnet provides 14 usable IP addresses, which is sufficient for all current devices. Seven usable IP addresses remain reserved for future expansion.